

Quantum Networks: A New Platform for Aerospace

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The ability to distribute entanglement between quantum nodes may unlock new capabilities in the future that include teleporting information across multinode networks, higher resolution detection via entangled sensor arrays, and measurements beyond the quantum limit enabled by networked atomic clocks. These new quantum networks also hold promise for the Aerospace community in areas such as deep space exploration, improved satellite communication, and synchronizing drone swarms. Although exciting, these applications are a long way off from providing a “real-world” benefit, as they have only been theoretically explored or demonstrated in small-scale experiments. An outstanding challenge is to identify near-term use cases for quantum networks; this may be an intriguing new area of interest for the aerospace community, as the quantum networking field would benefit from more multidisciplinary collaborations. This paper introduces quantum networking, discusses the difficulties in distributing entanglement within these networks, highlights recent progress toward this endeavor, and features two current case studies on mobile quantum nodes and an entangled clock network, both of which are relevant to the aerospace community.

I. Introduction

THE internet fundamentally changed how we live our lives. It not only allowed new ways to communicate, but also disrupted the status quo, from the music and television industries to the way society does business on a daily basis. Similarly, quantum networking stands to be an extremely disruptive quantum technology, as it will not only enable a new platform that offers theoretically secure communication protocols [1], but it will also provide a novel way to connect sensors and computers through entanglement distribution. These entanglement-assisted operations promise advanced capabilities that are impossible with classical technology [2–5] and may enable applications of interest to the aerospace community, such as novel time transfer protocols, quantum-enabled long-baseline interferometry, and distributed timing and sensing.

A quantum network is a collection of quantum devices connected via distributed entanglement, similar to a classical network but with new constraints imposed on the network by the requirement to maintain the “quantumness” of the information, as defined more precisely next. It is likely that quantum networking will progress in a similar manner to the related field of quantum computing. Many recent quantum-computing demonstrations are special-purpose devices with limited capabilities. These early devices help to inform researchers as the field moves toward the long-term goal of a fault-tolerant (or “universal”) device that can solve many more problems

than these early special-purpose models. Similarly, near-term quantum network demonstrations will operate over data-center or metropolitan-scale distances and have topologies designed for limited applications. The far-term goal is a universal network that connects many devices for more general purposes, similar to how the internet connects various devices for a myriad of functions [6]. We stress that quantum networking will not replace the internet as we know it (just as quantum computers will not replace our current computers) but will instead provide complementary functions that work in unison with these classical counterparts to provide new capabilities. We primarily focus on these long-distance networks in this paper to motivate a broader range of applications that will be possible in the future.

Of the three areas of Quantum Information Science (QIS), simulation and computing, sensing and metrology, and communication and networking, quantum networking is arguably the least mature (even though some of the early quantum key-distribution protocols predated any quantum computing implementations). This is due to the difficult challenge of creating, maintaining, and controlling entanglement for distribution beyond a single quantum module. Furthermore, for the universal network described earlier, one will need both transmission quantum bits (qubits) to send the information between the nodes, and stationary (likely matter-based) qubits to perform operations at the nodes and have the ability to seamlessly transfer information between the two. Unlike classical communication that uses millions of photons to transmit signals, quantum networking requires users to send and receive single photons. These single photons encode quantum states that must be coherently maintained during the networking or communication protocol. As we will discuss next, this need for quantum-encoded single photons makes quantum networking a much more challenging task than classical communication, as many of the tools and frequency ranges available for classical communication cannot be used. For example, room-temperature quantum communication is not possible in the radio-frequency or microwave part of the electromagnetic spectrum because the single-photon states will be overwhelmed by background blackbody radiation. Therefore, quantum networking experiments primarily use visible or infrared frequencies, and researchers have made substantial progress in sending single and

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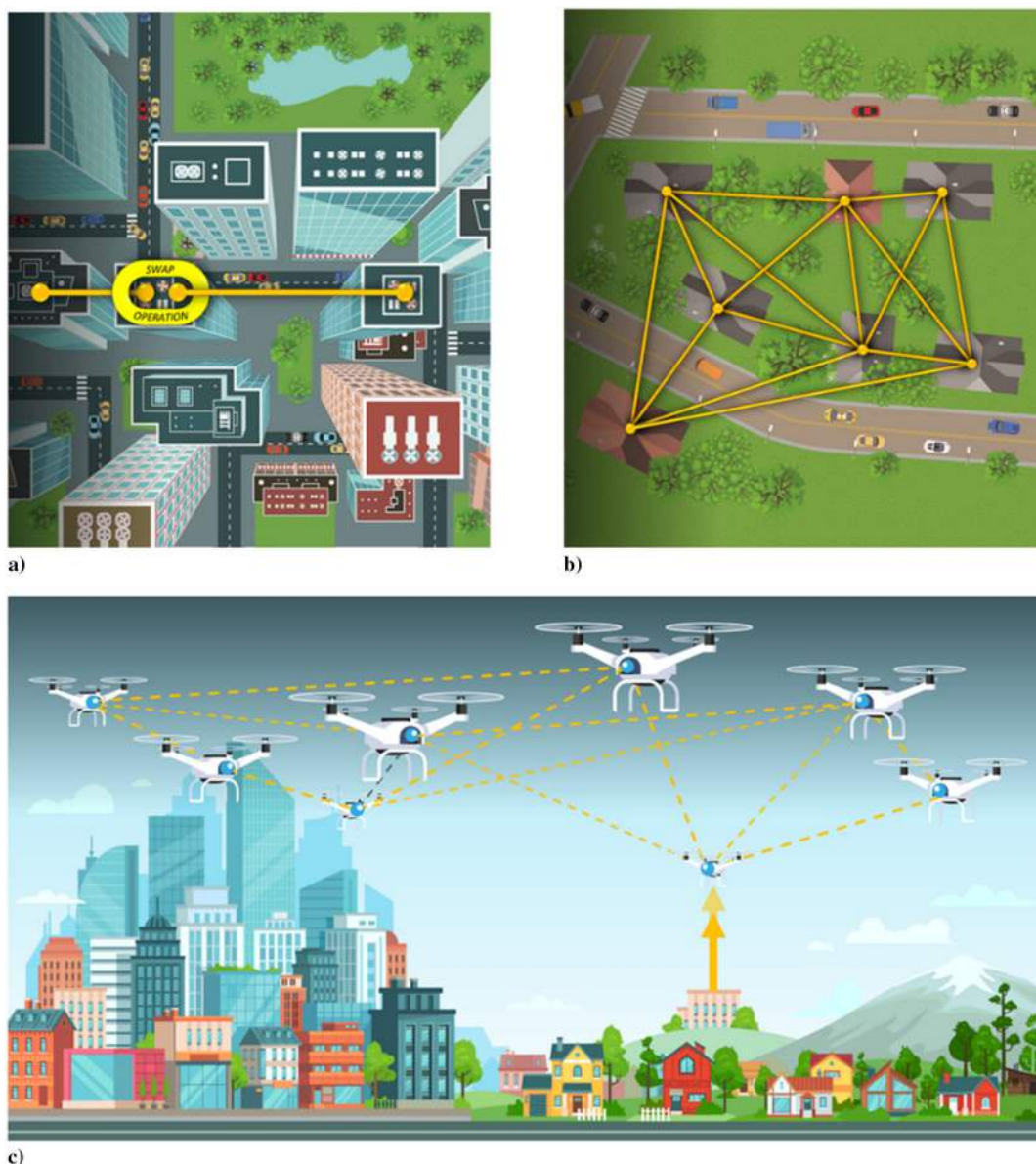


Fig. 1 Different types of quantum network configurations, including a) linear point-to-point network, b) nonlinear quantum network configuration, and c) a quantum mobile network.

entangled photons encoded with quantum information over long distances [7,8] and performing preliminary networking operations over short distances [9]. Because most matter-based qubits do not operate at common telecommunication frequencies, this necessitates a way to transduce, or frequency convert, the quantum information between the qubits at the nodes with the communication channel. Transduction is one of the biggest challenges in quantum networking, and although progress is being made in getting information in and out of the nodes using the intrinsic matter-based qubit frequencies [10–17], researchers have yet to demonstrate an efficient and effective transduction mechanism.

Another challenge in quantum networks is the fragility of entanglement, which leads to a simultaneous need to shield the quantum systems from unwanted noise while maintaining a connection to the environment to send the information between the nodes. In classical networks, repeaters amplify the signals during transmission. Unfortunately, one cannot directly replicate this in a quantum network, as one cannot copy or amplify quantum particles, due to the “no-cloning” theorem in quantum mechanics [1]. Therefore, one must operate repeaters differently in a quantum network, incorporating ancillary qubits and error-correction methods to restore the targeted quantum state before transmitting the information to another

node [18,19]. Another current outstanding experimental challenge to quantum networking is realizing a quantum repeater.

Figure 1 illustrates three possible quantum network configurations. Figure 1A is a point-to-point network that illustrates two end-points within a city connected by a series of quantum repeaters (only one shown in the figure). Here, quantum repeaters between the end nodes reduce the effect of loss, allowing one to transmit the information over longer distances [18]. Point-to-point networks are the most common quantum networking system developed to date and repeater-enhanced networks are in the early stages of development. Figure 1B is a multi-user network connecting points within a small cluster. This type of topology could connect nodes for distributed quantum computing or other local networking protocols. The figure shows a “partial mesh” architecture in which each node is connected to several (but not all) others,[†] but ultimately the end application will dictate how the modules are connected and what types of qubits reside within each node. Figure 1C illustrates a network of unmanned

[†]Other common networking topologies are a ring (each node connected to exactly two other nodes), star (all nodes but one connected to that one), and mesh (all-to-all connectivity); there are obviously many hybrid variants to these base topologies.

aerial systems (UAS) connected by distributed entanglement. Here, the entanglement is transferred to a UAS from a central ground node and then distributed among the other UASs. One can envision replacing any of the aerial nodes with a fixed (building) or mobile ground-level node (car, truck, train, ship, etc.). The multi-user networks shown in Figs. 1B and 1C are still largely conceptual.

Quantum networks may enable novel networking schemes, such as the ability to teleport quantum information between quantum nodes. Early laboratory demonstrations have teleported an unknown quantum state between two nodes [9,20,21], and there have been numerous proof-of-principle quantum network demonstrations from city [22,23], to intercity [24], to satellite scales [25]. In the very long term, quantum networks may provide theoretically secure communication protocols that would make eavesdropping nearly impossible. Important for some adversarial scenarios, direct optical links can be much more resilient to “jamming” than RF communications, which can be thwarted without an adversary knowing the path of the signal. However, for the same reason, direct optical channels are *more* susceptible to a different denial-of-service attack if they rely on a single-link implementation, blocking that optical link; expanding to multi-user networks reduces this denial-of-service vulnerability. A quantum network can also provide benefits to the fields of quantum sensing and quantum computing. For example, connecting quantum sensors through an entangled network may provide improved timing for clock synchronization, or better-resolution telescoping through protocols like quantum-enabled very-long-baseline interferometry [2]; a network can also connect modules of quantum processors and allow distributed quantum computing with greatly enhanced processing power [26].

This introduction offers far-reaching applications that future quantum networks may enable. However, there are still many challenges to extend entanglement beyond a few nodes, due to the intricate nature of the quantum systems and the difficulties in connecting disparate qubit technologies. The remaining sections offer a realistic view of what is currently possible, describe the outstanding challenges to realizing full (universal) quantum network capability, highlight recent initiatives, and highlight two use cases of interest to the aerospace community. Section II introduces quantum networking and defines important terms, Sec. III discusses the current status of quantum networks and outstanding challenges, Sec. IV provides case studies on quantum systems on an unmanned aerial system and a recent experiment entangling two atomic clocks, and Sec. V concludes with a summary.

II. Primer on Quantum Information Science and Quantum Networking

Quantum information science merges the established fields of quantum mechanics and information science in unique ways that bring novel fundamental insights to both disciplines while providing a foundation for entirely new theories and technologies. It exploits the laws of physics to realize unique physical systems that outperform their classical counterparts for select quantum networking and quantum computing use-cases [1]. This section provides a brief introduction to quantum networking and discusses the basic quantum-information-processing building blocks that serve as the foundation for these networks [27].

A. Quantum vs Classical Networking

A fundamental difference between classical and quantum networking is the use of single photons encoded with quantum information to transmit messages. Due to the extreme difficulty of operating at the single-photon level while maintaining quantum coherence, the frequency ranges and tool sets are limited as compared to what can be done in classical communication; for example, the no-cloning theorem of QIS prevents the use of in-line amplifiers to combat signal loss. Quantum networking protocols often use optical wavelengths, typically 800 and 1550 nm, where there may be less loss (e.g., loss in optical fibers has a minimum of ~ 0.15 dB/km around 1550 nm), as well as tolerable background, such as from blackbody radiation, in the environment at these wavelengths. Also, for communication in

the Earth’s atmosphere, there are spectral windows with lower loss from molecular absorption. Finally, the wavelength choice is often driven by the availability of high-efficiency, low-noise single-photon-counting detectors, such as silicon avalanche photodiodes that operate in the visible part of the spectrum, or classical communication equipment operating around 1550 nm that can be adapted for quantum networking systems. The blackbody radiation background, on the other hand, makes using microwave (or terahertz or radiofrequency) signals at room temperature very difficult, if not impossible, as there is no way to distinguish the signal from the large background or prevent interference from this large background. The qubit technologies that do use microwaves for communication protocols are limited to operating entirely within a cryogenic environment, with temperatures typically around 10 mK to reduce the background, so that one can distinguish between the qubit signals of interest and the background microwave environment.

Classical repeaters amplify signals between nodes to prevent information loss over long distances. Trying to use the same technique in a quantum network results in the amplification process destroying the quantum information, and so instead, one must use a much more complicated “quantum repeater,” which will be discussed later in the paper. A feature of quantum mechanics that further complicates operations is that measurement destroys the quantum properties encoded in a qubit and effectively “collapses” the encoded quantum information into a single (classical) outcome. It is not possible to reencode the resulting classical signal back into the (unknown) quantum signal and complete the protocol. Once the system is measured, the protocol is over, and one needs to start again. ** It is important to note that an operator can measure a system intentionally, but the environment can also effectively make a measurement of the system (the environment itself becoming entangled with the system) with or without the operator’s knowledge. Therefore, care must be taken to adequately shield or monitor the quantum channel to ensure that only intentional measurements are made.

Finally, as alluded to earlier, quantum mechanics does not allow one to make copies of an unknown quantum state [1]. These unique quantum mechanical properties make operating quantum systems for real-world applications challenging. However, they are also at the heart of the promising advances quantum information science has to offer, e.g., the no-cloning limitation is critical for the security of quantum cryptography.

B. Qubits

A qubit is the fundamental unit in quantum information and can be nearly any well-controlled, two-level quantum system. Some common qubit technologies include trapped ions, superconducting qubits, nitrogen vacancy centers in diamond, quantum dots, neutral atoms, and photons [28]. Unlike classical bits that can store only a “0” or a “1,” a qubit can store any arbitrary superposition of 0 and 1 at the same time. The ability to store and act on superposition states is a powerful feature in quantum information processing that allows one to simultaneously store and process 2^n states, where n is the number of qubits. When combined with the quantum mechanical phenomena of entanglement discussed next, these two properties allow one to fully harness quantum systems and provide unique capabilities compared to their classical counterparts. In some cases, it allows one to outperform classical systems, such as quantum computers that factor large numbers exponentially faster than any known classical algorithm [4] or that use quantum communication to generate random keys for theoretically secure protocols that are, in principal, secure against any attacks allowed by physical laws [3,29].

Entanglement is a special property shared among multiple quantum systems such that each subsystem does not have a definite value for one or more properties (like energy, or polarization) on its own: All the information is stored in the joint state of the particles, no

**Note that for quantum computing, one *does* make measurements on *some* of the qubits throughout a quantum calculation, in order to detect and correct errors; however, these must then be replaced with “fresh” ancilla qubits. Similar methods occur for error correction and entanglement distillation in quantum repeaters.

matter their separation. This leads to correlations among the outcomes of measurements on the particles, and these correlations exceed that which is possible with any nonquantum theory. It is important to note, this does not allow faster-than-light communication, as the protocols always require a classical (and therefore causal/nonsuperluminal) signal to facilitate the quantum information transfer. For quantum networking applications, this classical information does not reveal information about the message itself, but is auxiliary information about how to process the qubit to ensure the quantum protocol proceeds as expected; therefore, the sender can broadcast this classical information openly, as it does not provide insight or access to the message transmitting over the quantum channel.

C. Single- and Multiqubit Operations

Another necessity for quantum information processing and quantum networking is a set of universal logic gates. In classical computing, a single-bit gate performs a complete bit flip ($0 \leftrightarrow 1$); in contrast, a quantum single-qubit gate can also perform a full bit flip or can prepare any arbitrary superposition of 0 and 1, with an infinite number of possibilities. Two-qubit gates can be used to create entanglement, for example, a “target” qubit is flipped from $0 \rightarrow 1$ if and only if the “control” qubit is a 1. (Operating this “controlled not” gate with the control in a 0- and 1-superposition will entangle the two qubits.) For qubits within the same node (intranode), one implements these gates by coupling qubits in a specific way and the physical implementation is different for each qubit technology. For example, one can entangle trapped ions by coupling the spins of two ions through the phonon (or shared motional modes) of the trapped ion crystal [30,31]. For photons, entangled pairs can be created directly using nonlinear optical crystals, which convert one high-energy photon into two daughter photons entangled in some (or multiple) property.

Entanglement is commonly benchmarked using the metric of fidelity. Fidelity is a measure of the overlap between the targeted end state and the state actually created [1]. Single- and two-qubit gates are now routine laboratory operations, but the control of large quantum systems (>50 qubits) remains challenging, due to the unavoidable coupling to the environment. Coherence is a measure of how long a quantum state can maintain a superposition before a user intentionally acts on it (to make a measurement) or the environment perturbs it. Decoherence is anything that disturbs the quantum state and causes the system to lose coherence. Decoherence can take various forms that depend on the qubit technology and its surrounding environment; these include photons impinging on atoms, fluctuating magnetic fields altering the precession rates of spin qubits, dispersive birefringence in fiber dephasing polarization qubits, and many other effects.

Finally, quantum networking requires faithful quantum information transfer between distant nodes. Photon-mediated interactions are the main mechanism to realize connections between remote nodes, cf. recent examples using trapped ions [32] and nitrogen-vacancy centers in diamond [33], although multiple qubit technologies have demonstrated this type of operation, as discussed in the current paper and the references therein [34]. Both fiber and free-space channels (“wired” and “wireless” communication) are likely to be necessary for a longer-term quantum network because of loss considerations. Optical fiber has loss scaling exponentially with distance (e.g., -0.2 dB/km at 1550 nm), whereas long-distance free-space channels suffer loss from diffraction (transmission falls as $1/L^2$), and from turbulence if operating in the Earth’s atmosphere. Fully global quantum networking will likely require a space-based communication link, and this may be the lowest-loss solution.

III. Quantum Networking

A quantum network is a set of interconnected nodes linked via entanglement distribution, and protocols consist of applying quantum logic gates (single- and multiqubit), transmitting photons between the nodes, and making measurements. In principle, these networks can range from chip-scale to global-sized networks. The very long-term vision is a universal network with plug-and-play components that can operate over both short and long distances.

However, until researchers overcome the challenges needed to realize these longer-term networks, one must optimize current networking efforts for the specific end goals, as that will determine the number, the type of qubits, and the topology of the network. As we will discuss, most current network demonstrations are limited to chip-scale and local area networks due to outstanding challenges impeding progress toward long-range (wide-area and global-scale) quantum networks. We primarily focus the remaining discussion on networks using photons to transmit information over short and long distances.

A quantum network will allow novel networking schemes such as teleportation of quantum states [35]. Here, through the distribution of entanglement, single-qubit operations, and two-qubit measurements, quantum information can be teleported between remote parties. The allure of this protocol is that the information transfers over a theoretically secure quantum channel that is both tamper resistant and tamper evident. Although one always needs an associated classical channel to perform the quantum protocols, no pertinent information regarding the quantum message is ever sent over these channels, i.e., classical information can be widely broadcast without revealing the quantum message that is the state being teleported.

Beyond communications, quantum networks will also advance the other quantum-information-science areas of metrology, sensing, and computing. For example, global-scale distributed entanglement that connects remote atomic clock nodes will allow more precise timing [5]. Similarly, the ability to connect a network of telescopes via optical-spectrum entangled photons will allow higher resolution sensor protocols such as quantum-enabled very-long-baseline interferometry (VLBI) [2], which have a baseline that greatly exceeds what is currently possible using classical optical methods.^{††} In quantum computing, there is a practical limit to how many qubits can be controlled and operated within a single module, so the ability to connect remote modules will allow researchers to scale quantum computers beyond these limits [26,34,36]. Distributed quantum states will also allow researchers to securely send computational problems to and from quantum computers while maintaining information security and integrity through protocols such as “blind” quantum computing [37], realizing large-scale entanglement distribution will almost certainly unlock other yet-to-be-determined disruptive information sharing and networking capabilities.

A. Quantum Key Distribution

Many early quantum communication experiments focused on Quantum Key Distribution (QKD) and related quantum communication protocols. QKD allows remote users to securely generate a random key, a critical resource in various cryptography schemes like one-time pad encryption, by transmitting single or entangled photons [3,29]. Using quantum superpositions, e.g., of photon polarization or time bin, provides a physical layer of security because any information-revealing intrusions by a potential eavesdropper will necessarily lead to unavoidable and detectable errors, so that an insecure key is never used. However, making these QKD systems in practice is challenging, and demonstrations have suffered from imperfect implementations [38]; QKD thus serves as an interesting initial proving ground for a more general quantum network, as the protocol execution requires many of the essential networking ingredients discussed earlier. It should be noted that QKD is not currently recommended by the U.S. Government for deployment [39], although there are currently large QKD efforts elsewhere, e.g., China, Europe, Japan, and South Korea.

B. Quantum-Matter-Based Qubit Nodes and Quantum Interconnects

Photons are a mature qubit technology for quantum information science, especially when transferring quantum information at distances longer than a chip-scale device. Researchers have made remarkable progress in sending single photons over long distances,

^{††}It is true that there are existing VLBI systems, e.g., the Center for High Angular Resolution Astronomy array (see Ref. [33] here <https://www.chara.gsu.edu/astronomers/journal-articles> for more info), but channel loss limits the separations to hundreds of meters.

but these are largely proof-of-concept demonstrations that are not yet mature enough for deployment in critical systems. Examples of recent long-distance proof-of-concept quantum networks include transmitting entangled photons over 143 km between the Canary Islands in a free-space demonstration [7], transmitting single photons over 600 km in a fiber demonstration [8], and achieving conditional teleportation using telecommunication time-bin photonic qubits that were teleported over 22 km and then transmitted over an additional 22-km link to show future network viability [40]. The 2022 Nobel Prize in Physics was awarded to John Clauser, Alain Aspect, and Anton Zeilinger for entangled photon experiments. Photonic systems have also achieved the highest degrees of entanglement, including hyperentangled states, where one can store quantum information in multiple, simultaneous degrees of freedom [41,42], and are a potentially competitive technology toward building a quantum computer [9]. Space-based demonstrations include China's 2016 Micius satellite launch [25] (as part of the Quantum Experiments at Space Scale Mission), which was used to demonstrate ground-to-satellite QKD [43]; a long-distance QKD experiment between China and Austria via a satellite relay [44]; teleportation [45]; and distributed entanglement between two ground stations separated by 1200 km [46] {see ref [47] for a review of other recent satellite quantum communication and networking demonstrations}. One collaboration opportunity between the aerospace and quantum information communities is to mature these technologies, keeping in mind the lessons learned over the past decades in developing classical networks.

The challenge now lies in adding matter-based qubits to the network nodes. The National Science Foundation recently held a workshop on quantum interconnects, and the resulting report, "Development of Quantum Interconnects (QuICS) for Next-Generation Information Technologies," discusses the challenges toward developing these photon-matter qubit interconnects [48]. As the report highlights, quantum interconnects are paramount, as they will be a key component for nearly all future quantum-information-science technologies, from quantum sensors to full-scale quantum computers. For networks, the challenge is to convert the quantum information leaving the network nodes to a wavelength optimized for the specific network channel, and this will depend on whether that segment of the network is operating over fiber or free space. For example, even though many atomic systems naturally emit photons, these are rarely at the correct wavelength for common communication channels [1550 nm for fiber optic channels and typically between 750 and 1550 nm for free space (shorter wavelengths experience less diffraction, but background levels may be higher)], so one must quantum- frequency convert (or "transduce") the signal to the proper frequency range [49]. Although research is promising, many open questions remain, such as how to incorporate the memory components into an existing fiber infrastructure, e.g., most quantum memory options are relatively narrowband (10–100 MHz), not directly compatible with broadband entangled photons that may be spectrally multiplexed over 10 or 100 s of nanometers, i.e., many terahertz.

Extending these concepts, a universal network will likely require many different types of matter-based qubits (optimized for computing, sensing, etc.) interfaced to each other and to photons to achieve the long-term objectives discussed in this paper. Such heterogeneity is likely to be a requirement because each system has different strengths and weaknesses, such as long coherence time but slow operation or the opposite. A critical challenge is to interface these disparate qubit technologies to seamlessly transfer quantum information and entanglement across the network. Researchers have made tremendous progress understanding, creating, and manipulating entanglement within various matter-based qubits (intranode connections) [28], and preliminary two-node demonstrations [13,50,51] paved the way for entanglement distribution among three network nodes [33]. These proof-of-concept demonstrations provide the foundation to extend this work to larger networks; however, several outstanding obstacles stand in the way of increasing network sizes beyond a few nodes and realizing full quantum networking capability [52].

Full network operation also requires two-way communication at the nodes so that information can be read in and out. Most atom-based demonstrations to date have focused on sending a photon from the

atom, because inducing absorption in an atom from an incoming photon [53,54] is much more challenging than reading the photon out of the atom [55]. This problem becomes even more difficult for qubit technologies with no natural optical photonic output, such as superconducting qubits. There is much ongoing research for microwave-to-optical transduction [56] to connect remote superconducting qubit systems. Another active research area focuses on trying to connect matter-based qubits directly to each other, such as superconducting qubits to trapped ions or neutral atoms [57,58].

C. Quantum Repeaters

One area of intense research activity is realizing a quantum repeater [19]. There is a fundamental limit to how far a single photon can be transmitted before the signal is significantly attenuated. For example, a typical optical fiber has a minimum attenuation of -0.2 dB/km around 1550 nm, corresponding to only 1% transmission after 100 km and 10^{-20} after 1000 km! Free-space losses can be much less, but are limited by telescope diameters, atmospheric turbulence, and direct line-of-sight constraints. To enable quantum networking at practical rates over longer distances, one can use a repeater to break a long segment up into shorter ones, using quantum error correction and entanglement distillation techniques to restore the targeted quantum state before sending the information to a subsequent network node. Reference [18] presents three different repeater generations (1G, 2G, and 3G) and discusses how to overcome loss and operation errors for each generation. The authors provide a cost coefficient as an optimization parameter, as there is a trade space depending on operational parameters such as gate speed, gate fidelity, and coupling efficiency. To summarize, 3G quantum repeater architectures can achieve much higher speeds, but require much more complex quantum processing at the nodes and can tolerate less loss (at most, 50%) between them than 2G or 1G. At present, researchers are still working toward a first demonstration of a 1G quantum network.

Ultimately, which repeater generation works best for a specific outcome depends on the desired networking protocol one wants to implement, along with the qubit technologies and topologies available in the network. A quantum repeater has not yet been achieved; however, researchers are making progress. Using a single solid-state spin memory integrated in a nanophotonic diamond resonator, researchers were able to demonstrate entanglement distribution at rates higher than would be possible with direct transmission [59]. Other researchers recently demonstrated the first telecom-heralded entanglement between remote multimode solid-state quantum memories [60]; the entanglement was heralded by detection of a telecommunication photon at a rate of up to 1.4 kHz and stored for up to 25 μ s. In another recent paper [61], researchers demonstrated heralded entanglement distribution between two absorptive memories, generating atom-photon entanglement asynchronously and then performing an entanglement swap between two repeater segments once each atom-photon event was registered. An atomic rubidium ensemble served as a short-term memory while waiting for each atom-photon entanglement event to occur, resulting in a linear dependence on the entanglement generation rate vs a quadratic dependence when there is no memory present. Each of these results and their predecessors pave the way for a future long-distance quantum repeater demonstration; however, it should be stressed that the achieved communication rates in these tour-de-force proof-of-principle experiments are far below what would be necessary for any practical/useful applications. For example, the QKD 600-km fiber record of [8] achieved secret key rates of only 1 bit/s; in contrast, continuous one-time pad encoding of an audio signal, or blind remote quantum programming, might require >1 –1000 kbit/s.

D. Other Challenges

An outstanding nontrivial challenge is operating quantum networks in conjunction with classical network infrastructure. Many research groups have successfully operated quantum channels in parallel with classical channels, such as in Ref. [62], where researchers performed a fiber-based QKD experiment at 1550 nm with two

simultaneous 200-GHz-spaced classical channels in the same fiber. More recently, classical coexistence experiments have investigated the deleterious noise created by strong classical signals at 1550 nm on entangled photons at ~ 1310 nm, over a 45-km optical fiber link [63]; having the classical signal at longer wavelength (lower energy) reduces the noise from Raman-process photons. Another solution to the noise added by classical signals is to reduce those signals themselves, by using quantum receivers to detect them [64].

Another difficulty is operating quantum-information-science apparatuses outside of a well-controlled laboratory environment. Although many photon-based experiments have been successfully fielded, fewer matter-based qubits have been proven outside of a laboratory, and this is currently an active area of investigation. A promising example is the NASA Deep Space Atomic Clock that launched in 2019. This mission is testing the accuracy of a mercury-ion atomic clock in space; a recent publication detailing the clock performance since its launch [65] reports an order-of-magnitude performance improvement over space clocks currently in use. Another recent experiment demonstrated an elementary quantum link that connected two room-temperature quantum memories in nodes at Stony Brook University and Brookhaven National Laboratory, separated by 158 km [66]. Experiments like these pave the way for future fieldable distributed entanglement networks.

The final challenge we will discuss here is how to distribute quantum information and entanglement across more than two nodes. Only recently have researchers moved beyond two nodes and used pairwise entanglement to build up a three-node entangled state [33]. An active research area in quantum computing is how to create multipartite entanglement, how to operate experiments with high levels of entanglement, and how to verify and validate the entanglement. Quantum networking faces similar challenges but with the added complication of distributing this entanglement across remote nodes [67]. In Ref. [18], the authors compare three different protocols for connecting two quantum repeater nodes and discuss how one can combine these protocols to realize a larger network connectivity. Reference [68] presents a case where a global distributed entangled state is the starting point and targeted operations are performed within the network to generate a specific end state; the paper also examines multipartite vs bipartite entangled states and offers a quantum network configuration protocol to dynamically extend the network.

E. Quantum Network Initiatives

In much the same way that the release of IBM's 5-qubit "quantum computer" to the Cloud revolutionized the field of quantum computing [69], a distributed entanglement testing ground offers promise for quantum networking. The IBM 5-qubit cloud device offered researchers a platform to test concepts and develop a better understanding of how quantum computers operate; since then, multiple companies have made their own small-scale quantum processors accessible to the public. Quantum network testbeds, where one can investigate entanglement distribution and get a "hands-on" grasp of the challenges around large-scale entanglement distribution, will likely propel the field forward in a similar manner, and inspire new applications as more people learn what kinds of advantages quantum networks offer, and discover new capabilities {e.g., this is a major goal in the recent establishment of a permanent public-access quantum network node^{**}}. In addition, given the overwhelming amount of research still necessary to realize a quantum network, no one group can do it alone; therefore, we must strengthen ties across the many research groups working on quantum networking and come together. Such efforts are starting to take shape in many forms. For example, in 2013, the United Kingdom established several quantum hubs, one of which focuses on Communications and another on Networked Quantum Information Technologies [70], and in the European Union initiated a Quantum Flagship with one focus area on Quantum Communication [71]. As mentioned earlier, China has an effort to explore Quantum Experiments at Space Scale [72] and is working toward hybrid networks combining space- and fiber-based links [24].

The United States recently released the report, "A Coordinated Approach to Quantum Networking Research" [73], which details the ongoing activities within the U.S. government agencies that include multiple quantum networking testbeds currently under development.

IV. Case Studies

This section will present two quantum-networking case studies of interest to the aerospace community. The first will focus on entanglement distribution using an unmanned aerial system (UAS) and the challenges associated with operating outside of a laboratory environment. The second will present an experimental demonstration of two entangled quantum clocks and discuss the potential applications that this new tool could enable.

A. Case Study 1: Quantum Networks on Mobile Platforms

In future quantum-enabled systems, quantum information processors and sensors will be interconnected via a quantum network; at present, with few exceptions [24], the network nodes are at fixed stations and photons are exchanged over dedicated optical fibers or over free-space channels with fixed optical systems such as telescopes. On the other hand, there are many advantages to developing quantum networks on mobile platforms, such as those shown in Fig. 2, to realize flexible and easily reconfigurable quantum networking or sensor systems. For example, there are scenarios where fiber links are simply impossible (ships, planes, etc.), or where a fixed telescope is undesirable, e.g., direct line-of-sight over distances exceeding 10–100 km requires transmitter/receiver heights greater than 100 m to avoid occlusion by the (curved) Earth and to minimize the deleterious effects of turbulence (which drops off rapidly with altitude). As a second example, quantum-enhanced sensors will likely be optimally deployed *relative to some region/object of interest*, so being able to readily adjust their location is critical. Such platforms could also be useful for addressing the "last mile" problem that often plagues classical communication systems; as an extreme example, soldiers on a battlefield accessing quantum information resources will need small size, weight, and power (SWaP) solutions. This case study discusses applications of a mobile quantum network, technical challenges for its realization, and a brief review of current progress.

As discussed earlier, quantum networks may enable new applications that go beyond QKD and include blind quantum computing, distributed quantum computing, and distributed quantum sensing. In these applications, there are use cases requiring links beyond a fiber-based network. For example, to establish a secure quantum link across transcontinental or even transoceanic distances, we will either need a fiber link with many robust quantum repeater nodes, or a satellite entanglement "bridge," which will enable higher rates by many orders of magnitude, because the transmission for a satellite bridge falls off at worst with distance squared, compared to exponential dropoffs in fiber.

One exciting area of quantum information science is distributed quantum sensing, which we discuss by contrasting with classical sensor networks. In a classical sensor network, multiple devices are used to transduce information from the world around us, such as the presence and location of people, local magnetic fields, or local weather conditions. Data from each sensor is combined to improve the signal-to-noise ratio (SNR) and to create a spatially distributed map. In contrast, a quantum sensor network will use distributed devices that have a shared quantum state via distributed entanglement to greatly improve the SNR. In rough terms, the SNR of a classical sensor network scales as the square root of the number of devices, whereas it can scale linearly for a quantum sensor network [74].

For many envisioned applications, having a quantum sensor network on mobile systems will be important for several practical reasons. For example, mapping the Earth's anomaly magnetic and gravitational fields or searching for tunnels and caves will make the most sense using airborne or space-based quantum assets. Furthermore, we can envision having a small quantum computer on a satellite

^{**}Data available online at <https://granger.illinois.edu/news/stories/60290>.

tolerate large temperature variations and mechanical vibration to go beyond what has already been demonstrated [75]. Photonic integrated circuits are being developed to meet these needs, but the requirements of extremely low photon loss and pure quantum state generation have yet to be met; also, these devices often operate in the telecommunication band of the electromagnetic spectrum, where high-quality and low-SWaP single-photon-counting detectors are not available. Realizing mobile quantum networks requires substantial engineering effort to exceed the current state-of-the-art of the corresponding classical devices (e.g., background noise from the classical Pointing and Tracking locking signals is much more problematic for quantum transmissions than classical, and the quantum systems generally require higher levels of temperature and vibration control), but there seem to be no in-physical limits to prevent such developments. Certainly, the field would benefit greatly from input from the aerospace community.

B. Case Study 2: An Entangled Clock Network Demonstration

A particularly interesting quantum networking application for the aerospace community may be a quantum network of distributed atomic clocks. Atomic clocks with optical transitions are our most precise tools to measure time and frequency [78–80]. Optical atomic clocks provide more accuracy and precision than their microwave counterparts; Ref. [78] demonstrated an aluminum-ion-based optical atomic clock, which has a fractional frequency uncertainty of less than 10^{-18} ; this clock will not gain or lose one second in 33 billion years! Optical clocks that are networked classically could enable time transfer of picosecond accuracy, a level that is not achievable with the GPS or microwave atomic clocks. Further, success in the NASA Deep Space Atomic Clock experiments paves the way for future onboard clocks that may enable deep-space exploration [65] or enable more effective communication between satellite swarms. Here, we discuss the potential benefits of using atomic clocks that are linked via a quantum network and present a recent demonstration of how this remote entanglement might enhance timekeeping applications.

Frequency comparisons between atoms in separate locations can be used to probe the space–time variation of fundamental constants [81], to improve geodesy [81–83], or to study the properties of dark matter [84,85]. However, measurements on independent clock systems are limited by the standard quantum limit (SQL). Nonclassical states, such as entangled states, enable measurements beyond the SQL [74,86,87] and can attain the Heisenberg limit, the ultimate precision allowed by quantum mechanics, whereby the measurement precision scales as $1/N$, where N is the number of entangled systems (a $\sqrt{N}$ enhancement over measurements with N nonentangled systems, i.e., the SQL). Such “quantum-enhanced” measurements have been used for gravitational wave sensing [88,89], searches for dark matter [90], and force sensing [91,92].

For atomic clocks, local entangling operations have been used to demonstrate enhancements for atoms within the same device [93–99]. Quantum networks can be used for enhanced metrology by distributing entanglement between remote atomic clocks; such entanglement can enable frequency comparisons to move beyond the SQL and reach precision floors faster, leading to the detection of previously undetectable signals or the study of phenomena on shorter timescales. Using a quantum network for such applications is demanding, as it requires high-fidelity remote entanglement to generate an enhancement. Furthermore, the time to generate entanglement needs to be significantly shorter than the probe durations of the atomic clocks [typically about 500 ms [79]] to minimize the effect on the measurement duty cycle.

Recently, Ref. [100] reported the demonstration of entanglement-enhanced frequency comparisons using an elementary quantum network consisting of two $^{88}\text{Sr}^+$ ions, separated by ~ 2 m, which were entangled using a photonic link [15,26]. Entanglement reduced the measurement uncertainty for frequency comparisons by approximately $\sqrt{2}$, as predicted for the Heisenberg limit, thus halving the number of measurements required to reach a given precision compared to independent measurements on each ion. In addition, entanglement can offer insensitivity to dephasing of the probe laser, which is a

limitation to the achievable probe durations, and hence the measurement uncertainty, with present optical clock technology [101,102]. Conventional correlation spectroscopy techniques [103–106], which involve simultaneous measurements of an unentangled two-atom state using a common probe laser [107], can enable interrogation times beyond the coherence time of the laser, as recently demonstrated in Ref. [108] for two macroscopically separated clocks. In this regime, compared to correlation spectroscopy, the experiment in Ref. [100] demonstrated that the use of entanglement decreased the measurement uncertainty by nearly a factor of 2, leading to a factor of 4 reduction in the number of measurements required. Thus, the use of entanglement combines insensitivity to probe laser phase noise with the maximum measurement precision permitted by quantum mechanics.

An overview of the experimental apparatus in Ref. [100] is shown in Fig. 3a. Remote entanglement is created between two $^{88}\text{Sr}^+$ ions that are separated by about 2 m. Using a photonic link, the authors achieved a remote entanglement fidelity of 0.960(2) in an average duration of 9 ms [109]. As shown in Fig. 3b, they observed a reduction in the measurement uncertainty for a frequency shift applied to one of the ions using entanglement-enhanced spectroscopy, compared to that obtained using correlation spectroscopy with unentangled atoms.

The achievable probe durations in Ref. [100] of about 10 ms were short compared to the limit set by the $^{88}\text{Sr}^+ 4D_{5/2}$ lifetime [110] of ~ 400 ms, due to qubit decoherence from magnetic field fluctuations. This decoherence limited the absolute measurement precision to a fractional frequency uncertainty of $\sim 10^{-15}$, approximately three orders of magnitude worse than the state of the art for optical clocks [78]. These magnetic field fluctuations could be reduced by using superconducting solenoids [111], mu-metal shielding [112], or more advanced dynamical decoupling schemes [113]. The remote entanglement could in principle be mapped to any other ion species via quantum logic operations [114], with negligible loss of fidelity or speed using local mixed-species entangling operations [115]. Longer

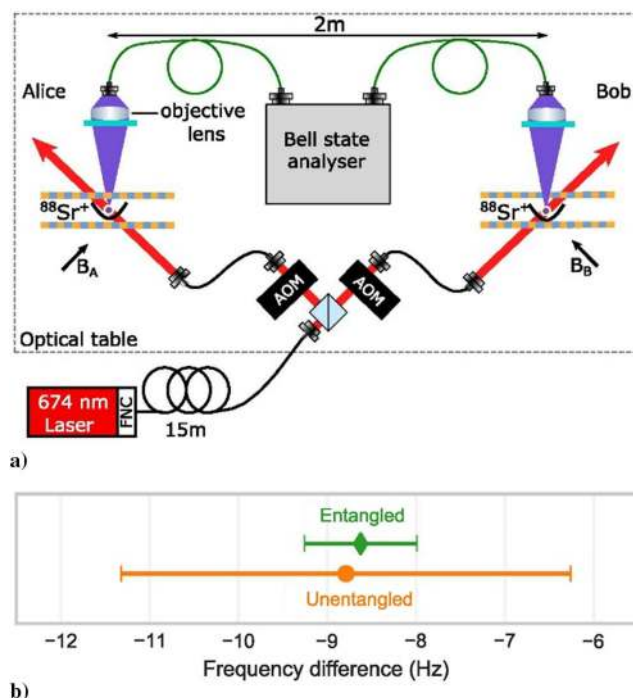


Fig. 3 a) Experimental apparatus for the initial demonstration of an elementary entangled clock network in Ref. [100]. b) Measurement of an applied frequency shift between the two clocks of -8.6 ± 0.6 Hz with (green) and -8.8 ± 2.5 Hz without (orange) entanglement on a 445 THz optical transition. With entanglement, they observed a much smaller uncertainty. {figure adapted from Ref. [100]}. (Adapted with permission from Nichol, B. C., Srinivas, R., Nadlinger, D. P., Drmota, P., Main, D., Aranedo, G., Ballance, C. J., and Lucas, D. M., “An Elementary Quantum Network of Entangled Optical Atomic Clocks,” *Nature*, Vol. 609, No. 7928, 2022, pp. 689–694, <https://doi.org/10.1038/s41586-022-05088-z>, 2022, Springer Nature.)

probe durations could be achieved with a secondary ion species with a transition that has a reduced magnetic field sensitivity [78] or a narrower linewidth. The secondary ion species could also have an increased sensitivity to fundamental constants [85].

For remote sensing applications of geodesy, larger distances between the clock nodes will be required. Although longer fibers with phase noise cancellation [116] could be used, the entanglement rate would be quickly reduced due to substantial fiber losses at 422 nm; this reduction could be potentially mitigated by frequency conversion to telecommunication wavelengths [117]. A larger network of atomic clocks, with more ions in each node, could reduce the measurement uncertainty further [5], either by having more ions in each node with local entangling operations, or with more nodes connected via additional photonic links.

V. Conclusions

The ability to distribute and harness quantum states will lead to exciting new capabilities for networking and communication, as outlined in this paper. Distributed entanglement will provide a fundamentally new way to communicate, through protocols such as teleportation, and may ultimately enable new capabilities, such as entangled sensor arrays, which will provide unprecedented imaging capabilities, new modalities for time transfer across networks, unparalleled accuracy for time keeping, the ability to perform distributed quantum computing, and other yet-to-be-imagined possibilities. Although researchers are making progress in understanding entanglement distribution, there are many outstanding challenges to overcome and many questions to answer. One of these is ascertaining some meaningful short-term use cases for these new networks. The aerospace community could play a pivotal role in unlocking some of these answers through increased collaboration and multidisciplinary opportunities. Once realized, this new tool set could provide novel solutions to current and future problems related to drone synchronization, satellite communications, and deep space exploration. Despite the difficulties ahead, the future holds great promise for this new networking paradigm.

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